

EEGNet Failure Analysis & Beyond

Defence presentation — PhysioNet EEGBCI motor-imagery subset

Slide 1 — The question and the constraints

EEGNet works in the lab. Where does it fail when we move it to a new person, and what does the failure tell us about how to fix it?

Train	Test	Task	Channels	Sampling	Trials/subject
Subj 1–8	Subj 9–10	Imagined fists vs feet (PhysioNet Task 4)	64	160 Hz	~45

Three seeds (1337, 2024, 7). Always report **best AND worst** — no cherry-picking. Source of truth: `results/metrics.json`.

Slide 2 — Three experiments to localise the failure

Setting	Train subjects	Test subjects
Cross-subject, full pool	1–8 (8 subjects)	9–10
Cross-subject, reduced pool	1, 2, 3 (3 subjects)	9–10
UMAP of the model's internal features	diagnostic	diagnostic

No within-subject baseline. The thing under test is **transfer to a new person**, not raw capacity. Within-subject EEGNet is already established as working (Lawhern 2018).

Slide 3 — Baseline results — the clue is in the diagnostic, not the accuracy

Setting	Best	Worst	Sil(class)	Sil(subject)
Full 8 subj	0.644	0.511	0.010	0.084
Reduced 3 subj	0.656	0.533	-0.002	0.062

Two things to notice:

1. Adding subjects did **NOT** help ($0.644 \approx 0.656$). One more training subject is worth roughly zero.
2. **Subject clustering** $\gg$ **class clustering** (0.084 vs 0.010). The network's main feature axis is "which person is this?", not "left or right?".

Sil(subject) = how clustered the model's features are by person.

Slide 4 — Why — the mechanism

EEGNet's spatial filter has shape `(filters, channels)` — one weight per (filter, electrode-index) pair. It looks at electrodes by *index*, not by *position on the head*.

Two structural facts:

1. The depthwise weight has **no notion of physical electrode position**. Move the cap 1 cm and the same weight reads a different brain region — but the network has no way to know.
2. Each person's electrical fingerprint (skull thickness, electrode contact, head shape) shows up as resting-state covariance with very high signal-to-noise. The classifier finds it because it is *easier to learn than the actual class signal*.

The high sil(subject) is what tells me that — the model basically

Slide 5 — Part 2 — TopoNet — two changes

5a. Per-trial Riemannian whitening

$$\tilde{X} = (XX^\top / T + \varepsilon I)^{-1/2} X$$

In plain English: rescale each trial so its covariance matrix becomes the identity. This mathematically removes the person-specific covariance fingerprint. No calibration data needed from the test subject — every trial gets whitened on its own.

5b. Coordinate-Conditioned Spatial Filters (CCSF)

$$W[c, k] = \text{MLP}([\text{pos}_c, e_k])$$

The spatial filter is no longer a lookup table. It's a small network that takes the actual 3D position of each electrode as input. Same physical position → same filter response, regardless of whose head

Slide 6 — TopoNet results — an honest split

Setting	Best	Worst	Sil(class)	Sil(subject)
Baseline, 8 subj	0.644	0.511	0.010	0.084
TopoNet, 8 subj	0.489	0.456	0.005	0.014
Baseline, 3 subj	0.656	0.533	-0.002	0.062
TopoNet, 3 subj	0.622	0.600	0.006	0.028

What's working and what isn't:

- **Mechanism works.** Subject clustering drops 6× on the full pool and 2.3× on the reduced pool — the Part 1 diagnostic moves a lot.
- **Full-pool accuracy drops.** TopoNet sits below the baseline here.
- **Reduced-pool accuracy improves and becomes stable.** Every seed lands in [0.60, 0.62]

Slide 7 — Why TopoNet wins on 3 subjects and loses on 8

The motor-imagery class signal for runs 6/10/14 lives in the **lateral-vs-medial covariance contrast** — bilateral hand area (C3/C4) for imagined fists vs midline foot area (Cz) for imagined feet. Per-trial whitening flattens covariance across the board, so it removes the class signal along with the person-specific component — they live in the same algebraic object.

- **Full pool:** EEGNet has enough data to find robust filters without the prior. Whitening becomes *cost without benefit*.
- **Reduced pool:** EEGNet was overfitting. Whitening's prior gives it a cleaner objective and stabilises every seed.

I threw the baby out with the bathwater on the full pool. The next fix would be **subject-conditional whitening** — remove only the

Slide 7b — Ablation decomposes TopoNet's two halves

`scripts/ablation_part2.py` :

Config	8-subj best/worst	3-subj best/worst	sil(subj) 8 / 3
Baseline	0.644 / 0.511	0.656 / 0.533	0.084 / 0.062
Whitening only	0.600 / 0.489	0.600 / 0.489	0.022 / 0.017
CCSF only	0.522 / 0.422	0.611 / 0.567	0.022 / 0.020
TopoNet (both)	0.489 / 0.456	0.622 / 0.600	0.014 / 0.028

Three findings:

1. **Either intervention alone cuts subject clustering $\sim 4\times$.** Both halves independently attack the mechanism from Part 1.
2. **They stack negatively on accuracy** — both remove some of the

Slide 8 — Part 3 — Hemispheric contrast channels

The novel idea. Half-page rationale in `report/part3_idea.md` .

Claim: the useful signal in EEG is in the *relationships* between electrodes, not in any one channel's amplitude. Volume conduction (skull and scalp spread each brain signal across many electrodes) means 64 channels are 64 overlapping views of a smaller set of sources. The relationships are already in the input, but buried.

Idea: add ten extra channels next to the raw 64. Nine are left-right symmetric pairs near motor cortex (C3 – C4, FC3 – FC4, CP3 – CP4 and neighbours). The tenth is a lateral-vs-midline contrast: $(C3 + C4)/2 - Cz$. Input becomes 74 channels.

Add, not replace. If the contrasts are useless the model can

Slide 9 — Why this is grounded, specific, and (I think) not just bipolar

Grounded. EEG has high temporal resolution but its spatial picture is coarse. Volume conduction means neighbouring channels end up carrying mostly the same thing. The useful information often lives in the differences between electrodes.

Specific. Ten exact contrasts, named electrodes, fixed subtraction. `scripts/plot_contrasts.py` plots the $C3 - C4$ and $(C3 + C4)/2 - Cz$ traces split by class — reviewers can verify the sign flip on $C3 - C4$ themselves.

Not just a bipolar montage. Bipolar, surface Laplacian, common-average, CSP, ICA — they all replace the raw signal with a fixed transform before the network ever sees the data. Once that's

Slide 10 — Sanity check on the engineered feature

`scripts/plot_contrasts.py` → `results/figures/part3_contrasts.png` .

Trial-averaged contrast traces, split by class:

- **C3 – C4 has a sign flip between classes** (T1 mean negative, T2 mean positive) — the lateralization pattern the contrast was built to pick up.
- Effect size is modest at the trial level (Cohen's $d \approx 0.14$ on C3 – C4, ≈ 0.05 on $(C3 + C4)/2 - Cz$). The contrasts are class-informative but **not classifiers on their own** — they are a useful representation for the network to lean on.

Slide 11 — Part 3 results — the prediction holds on the bigger pool

Setting	Best	Worst	Sil(class)	Sil(subject)
Baseline, 8 subj	0.644	0.511	0.010	0.084
EEGNet + contrasts, 8 subj	0.667	0.556	-0.001	0.037
Baseline, 3 subj	0.656	0.533	-0.002	0.062
EEGNet + contrasts, 3 subj	0.533	0.511	0.005	0.037

Full 8-subject pool — hypothesis held. Subject clustering drops $2.3\times$ ($0.084 \rightarrow 0.037$). Best accuracy up ($0.644 \rightarrow 0.667$). Worst case up ($0.511 \rightarrow 0.556$). Mechanism and outcome both lined up with what I'd predicted.

Reduced 3-subject pool — mechanism still works, but accuracy

Slide 11b — Did the model actually use the contrasts?

`scripts/part3_filter_analysis.py` → `results/figures/part3_filter_weights.png`.

I ran the test. The numbers came in mixed.

- Random baseline (every channel equally weighted): $10/74 \approx 0.135$.
- Trained mean per-filter share on contrasts: 0.150 — only $1.11\times$ the baseline. Small shift in aggregate.
- Per-filter spread: 0.031 to 0.263. Five filters at 0.20–0.26 (so about $1.5\text{--}2\times$ baseline). Five at 0.03–0.10, basically ignoring the contrasts. The other six are near the baseline.

So I'm walking the strong reading back. The aggregate number doesn't support "the model heavily leans on the prior overall" — I'd be overclaiming if I said that.

Slide 12 — Defence Q&A I expect

"Isn't this just a bipolar montage / surface Laplacian? Those are decades old."

Bipolar and Laplacian replace the raw signal with a fixed transform before the network sees anything — once it's done, the raw channels are gone. What I do is different: I leave the raw 64 in place and add 10 contrast channels next to them, and the depthwise filter decides per filter how much to weight each side. I ran the test on myself

(`scripts/part3_filter_analysis.py`) — the trained mean share on contrasts came back at 0.150 against a 0.135 random baseline, so only about 11 % above chance. I'm not going to claim the model heavily leans on the prior overall — the aggregate number doesn't back that. But the per-filter spread is wide (0.031 to 0.263):

Slide 13 — What I would build next

1. **Subject-conditional whitening** — remove only the subject-specific component of covariance, keep the class-discriminative asymmetry that simple whitening also removes.
2. **Frozen contrast weights on small pools** — freeze the depthwise filter weights on the 10 contrast channels in the reduced-data regime so the prior is preserved without paying the parameter cost.
3. **Pre-train CCSF MLP across multiple EEG datasets** — it's channel-count agnostic, so cross-corpus pre-training is free.

Slide 13b — Visual evidence — topomaps tell the same story

`results/figures/topomap_eegnet.png` — EEGNet's 16 depthwise filters.

Fragmented, noisy spatial profiles — no clean motor-cortex localisation. These are not motor-area detectors; they are person-specific covariance patterns.

`results/figures/topomap_ccsf.png` — CCSF's 16 filters.

Smooth gradients, all nearly identical. The MLP-on-position prior collapses to a small set of low-frequency spherical patterns. This explains CCSF's full-pool underperformance: not too little capacity, but too little *diversity*.

`results/figures/per_subject_accuracy.png` — EEGNet hits S10 (0.71) much better than S9 (0.58). The 0.644 headline number averages over a real asymmetry — subjects are not equally transferable.

Slide 13c — Classical reference — CSP + LDA

Setting	best / worst
CSP+LDA, 8 subj	0.489 / 0.489
CSP+LDA, 3 subj	0.367 / 0.367

CSP is **at chance** on the full pool and **below chance** on the reduced pool. CSP filters are explicitly person-specific covariance maximisers — under reduced-data cross-subject training they anti-align to test subjects.

It's the same failure mechanism Part 1 spelled out for EEGNet, showing up even more starkly in a model that's covariance-based by construction.

EEGNet (0.570 mean) uses temporal-frequency capacity that CSP

Slide 14 — Take-aways

- The cross-subject failure is **structural, not statistical**.
Adding subjects does not fix it ($0.644 \approx 0.656$). CSP+LDA at chance empirically confirms the mechanism.
- The diagnostic that reveals the mechanism is **subject clustering**, not accuracy. Accuracy on a 90-trial test set has a $\pm 5\%$ noise floor.
- **Part 2 ablation** isolated CCSF as the cleaner half — best worst-case test on the reduced pool (0.567). Topomaps show CCSF's next-step problem (filter collapse).
- **Part 3 hemispheric contrasts** validated the inductive-bias hypothesis on the full pool (subject clustering $\downarrow 2.3\times$, best test $\uparrow 0.644 \rightarrow 0.667$, worst $\uparrow 0.511 \rightarrow 0.556$). On the small pool the mechanism still works but parameter cost wins